

Ansible Interview Questions

50 multiple-choice questions covering the basics and core principles of Ansible, with answers and short explanations.

Section 1 - Ansible Basics & Architecture

Q1. Which statement best describes Ansible?

- A. A compiled programming language
- B. An agentless IT automation and configuration management tool
- C. A relational database engine
- D. A container runtime

Answer: B

Explanation: Ansible is an open-source, agentless automation tool used for configuration management, application deployment, and orchestration.

Q2. Which protocol does Ansible use by default to communicate with managed Linux nodes?

- A. FTP
- B. SSH
- C. Telnet
- D. SNMP

Answer: B

Explanation: Ansible connects to managed Linux/Unix nodes using SSH by default, which is why no agent is required on the target.

Q3. Ansible is described as 'agentless'. What does that mean?

- A. It runs without any controller machine
- B. No persistent software needs to be installed on managed nodes
- C. It does not require Python
- D. It avoids using SSH keys

Answer: B

Explanation: No long-running daemon is installed on the managed nodes; Ansible pushes modules over SSH (or WinRM) and executes them on demand.

Q4. Which language are most Ansible modules written in?

- A. Ruby
- B. Go
- C. Python
- D. Bash

Answer: C

Explanation: Most core Ansible modules are written in Python, although modules can be written in any language that returns JSON.

Q5. Which file format is primarily used to write Ansible playbooks?

- A. XML
- B. INI
- C. YAML
- D. TOML

Answer: C

Explanation: Ansible playbooks are written in YAML, which emphasizes human readability through indentation.

Q6. What is the default location of the main Ansible configuration file?

- A. /etc/ansible/ansible.cfg
- B. /var/lib/ansible/config
- C. /usr/local/ansible.cfg
- D. /root/.ansible

Answer: A

Explanation: The system-wide Ansible configuration file is /etc/ansible/ansible.cfg. It can be overridden by ANSIBLE_CONFIG, .ansible.cfg, or ~/.ansible.cfg.

Q7. Which component lists the hosts and groups that Ansible manages?

- A. Playbook
- B. Inventory
- C. Module
- D. Handler

Answer: B

Explanation: The inventory defines the hosts and groups that plays target. It can be static (INI/YAML) or dynamic (script/plugin).

Q8. Where is the default static inventory file located?

- A. /etc/ansible/hosts
- B. /etc/hosts
- C. /var/ansible/inventory
- D. /opt/ansible/hosts

Answer: A

Explanation: By default Ansible reads /etc/ansible/hosts. You can override it with -i or with the inventory key in ansible.cfg.

Q9. What does an Ansible 'play' represent?

- A. A single command run on a host
- B. A mapping of a group of hosts to a set of tasks
- C. A reusable unit of configuration like a role
- D. A connection plugin

Answer: B

Explanation: A play maps hosts (or groups) to an ordered list of tasks to be executed against them.

Q10. Which is the smallest unit of action in an Ansible playbook?

- A. Play
- B. Role
- C. Task
- D. Handler

Answer: C

Explanation: A task is a single call to an Ansible module and is the smallest executable unit inside a play.

Section 2 - Modules & Tasks

Q11. Which Ansible module is used to manage packages on Red Hat-based systems?

- A. apt
- B. yum
- C. pacman
- D. zypper

Answer: B

Explanation: yum (or dnf) manages packages on RHEL/CentOS/Fedora. apt is for Debian/Ubuntu.

Q12. Which module would you typically use to copy a file from the controller to a managed node?

- A. fetch
- B. copy
- C. synchronize
- D. template

Answer: B

Explanation: The copy module copies files from the control node to remote hosts. fetch goes the other direction.

Q13. What is the primary difference between the copy and template modules?

- A. copy is faster than template
- B. template renders Jinja2 variables before copying; copy does not
- C. template only works on Windows hosts
- D. copy can only handle binary files

Answer: B

Explanation: template processes the source file as a Jinja2 template, substituting variables, then transfers the rendered result.

Q14. Which module is used to manage services (start/stop/enable)?

- A. service
- B. package
- C. command
- D. shell

Answer: A

Explanation: The service (or systemd) module manages service state and enable-on-boot behavior.

Q15. Which module would you use to execute a shell command that requires shell features like pipes or redirection?

- A. command
- B. raw
- C. shell
- D. script

Answer: C

Explanation: The shell module runs commands through a shell, so it supports pipes, redirection, and shell variables. command does not.

Q16. Which module runs a command WITHOUT going through a shell?

- A. shell
- B. command
- C. raw
- D. expect

Answer: B

Explanation: The command module executes the binary directly, so shell operators like |, <, > are treated as literal characters.

Q17. Which Ansible concept makes most modules safe to run repeatedly with the same result?

- A. Concurrency
- B. Idempotency
- C. Reentrancy
- D. Parallelism

Answer: B

Explanation: Idempotency means a task only changes state when required, so repeated runs converge to the desired state without side effects.

Q18. What does the 'changed' status of a task indicate?

- A. The task failed
- B. The module modified the target system to reach the desired state
- C. The task was skipped
- D. The task is deprecated

Answer: B

Explanation: 'changed' means the module took action because the system was not already in the desired state.

Q19. Which module is best suited for editing single lines in an existing file?

- A. copy
- B. lineinfile
- C. template
- D. blockinfile

Answer: B

Explanation: lineinfile ensures a particular line exists (or is replaced) in a file. blockinfile manages multi-line blocks.

Q20. Which module fetches files FROM remote hosts back to the controller?

- A. copy
- B. synchronize
- C. fetch
- D. get_url

Answer: C

Explanation: fetch retrieves files from managed hosts and stores them on the control node, typically organized per host.

Section 3 - Variables, Facts & Templates

Q21. Which Jinja2 syntax is used to reference a variable in a template?

- A. \${var}
- B. {{ var }}
- C. <% var %>
- D. #{var}

Answer: B

Explanation: Jinja2 uses double curly braces for expressions: {{ variable }}.

Q22. What are 'facts' in Ansible?

- A. User-defined variables in vars files
- B. Information automatically discovered about managed hosts
- C. Static playbook constants
- D. Encrypted secrets

Answer: B

Explanation: Facts are data gathered from managed hosts (OS, IPs, memory, etc.) by the setup module at the start of a play.

Q23. Which module gathers facts about managed hosts?

- A. gather
- B. facts
- C. setup
- D. info

Answer: C

Explanation: The setup module collects host facts. It is invoked automatically unless gather_facts: false is set.

Q24. How do you disable automatic fact gathering for a play?

- A. facts: off
- B. gather_facts: false
- C. no_facts: true
- D. setup: skip

Answer: B

Explanation: Set gather_facts: false at the play level to skip running the setup module.

Q25. Which has the HIGHEST precedence among these variable sources?

- A. Role defaults
- B. Inventory group_vars
- C. Extra vars passed with -e
- D. Playbook vars

Answer: C

Explanation: Extra variables (-e / --extra-vars) have the highest precedence in Ansible's variable precedence order.

Q26. Which directive captures the output of a task into a variable?

- A. save
- B. register
- C. set_fact
- D. capture

Answer: B

Explanation: register stores the task's result in the named variable for use in subsequent tasks.

Q27. Which module/directive sets a variable dynamically during a play?

- A. vars_prompt
- B. set_fact
- C. include_vars
- D. register

Answer: B

Explanation: set_fact creates new variables at runtime that persist for the remainder of the play (and can be made host-level facts).

Q28. Which Jinja2 filter provides a default value if a variable is undefined?

- A. fallback
- B. default
- C. or
- D. ifnull

Answer: B

Explanation: The default filter, e.g. `{{ my_var | default('value') }}`, is the idiomatic way to supply a fallback.

Q29. Where should you place files that should be rendered by the template module in a role?

- A. files/
- B. templates/
- C. vars/
- D. tasks/

Answer: B

Explanation: The template module looks in the role's templates/ directory for source files (typically *.j2).

Q30. Which directory in a role holds static files used by the copy module?

- A. files/
- B. templates/
- C. defaults/
- D. meta/

Answer: A

Explanation: The files/ directory in a role holds static content referenced by the copy and script modules.

Section 4 - Control Flow, Handlers & Execution

Q31. Which keyword applies a condition to a task?

- A. if
- B. when
- C. only_if
- D. condition

Answer: B

Explanation: Use 'when:' followed by a Jinja2 expression to conditionally run a task.

Q32. Which keyword iterates a task over a list of items?

- A. for
- B. loop
- C. each
- D. iterate

Answer: B

Explanation: loop: is the modern keyword for iteration; with_items is the older equivalent.

Q33. What is a 'handler' in Ansible?

- A. A special task that runs only when notified by another task
- B. An alias for a role
- C. A connection plugin
- D. A type of inventory

Answer: A

Explanation: Handlers are tasks invoked by 'notify'; they typically run once at the end of a play, e.g., to restart a service after a change.

Q34. Which keyword is used to trigger a handler from a task?

- A. call
- B. trigger
- C. notify
- D. invoke

Answer: C

Explanation: A task uses 'notify: ' to schedule that handler for execution at the end of the play.

Q35. Which keyword causes a play to keep running tasks even after a failure on a host?

- A. ignore_errors: true
- B. continue: true
- C. no_fail: true
- D. failed_when: false

Answer: A

Explanation: ignore_errors: true makes Ansible record the failure but continue with subsequent tasks for that host.

Q36. Which directive limits how many hosts run a play in parallel batches?

- A. forks
- B. serial
- C. throttle
- D. batch

Answer: B

Explanation: serial controls rolling-update batch size at the play level. forks is a global parallelism setting.

Q37. Which keyword runs a task only once across all targeted hosts?

- A. once: true
- B. run_once: true
- C. delegate_to: localhost
- D. single: true

Answer: B

Explanation: run_once: true ensures the task executes on only one host (the first in the batch) regardless of how many are targeted.

Q38. Which directive runs a task on a host different from the current target?

- A. delegate_to
- B. redirect_to
- C. run_on
- D. host_to

Answer: A

Explanation: delegate_to redirects task execution to a specified host while still being part of the current play.

Q39. Which keyword tags tasks so they can be selectively run or skipped?

- A. labels
- B. tags
- C. marks
- D. scopes

Answer: B

Explanation: tags let you run subsets of a playbook with --tags or skip them with --skip-tags.

Q40. Which command initializes a new role skeleton?

- A. ansible-playbook init role
- B. ansible-galaxy init
- C. ansible-galaxy role init
- D. ansible role new

Answer: C

Explanation: Modern syntax: ansible-galaxy role init . The legacy shortcut 'ansible-galaxy init' also still works.

Section 5 - Roles, Collections, Vault & Tooling

Q41. Which directory in a role holds the default variables (lowest precedence)?

- A. vars/
- B. defaults/
- C. meta/
- D. vars_files/

Answer: B

Explanation: defaults/main.yml has the lowest precedence so consumers can easily override them; vars/ has higher precedence.

Q42. Which file declares a role's dependencies on other roles?

- A. meta/main.yml
- B. tasks/main.yml
- C. defaults/main.yml
- D. handlers/main.yml

Answer: A

Explanation: meta/main.yml lists dependencies, supported platforms, and Galaxy metadata for a role.

Q43. What is an Ansible Collection?

- A. A bundle of related roles, modules, plugins, and playbooks distributed together
- B. A group of inventory hosts
- C. A snapshot of facts
- D. A type of vault file

Answer: A

Explanation: Collections package and version content (modules, roles, plugins) and are distributed via Ansible Galaxy or a private hub.

Q44. Which command installs a collection from Ansible Galaxy?

- A. ansible-galaxy install collection
- B. ansible-galaxy collection install
- C. ansible install-collection
- D. ansible-collection get

Answer: B

Explanation: Use: ansible-galaxy collection install namespace.collection.

Q45. What is Ansible Vault used for?

- A. Storing inventory at scale
- B. Encrypting sensitive data such as passwords or keys in playbooks
- C. Caching facts between runs
- D. Logging task output securely

Answer: B

Explanation: Ansible Vault encrypts variables and files so secrets can be safely committed to source control.

Q46. Which command encrypts an existing plain-text file with Ansible Vault?

- A. ansible-vault encrypt
- B. ansible vault lock
- C. ansible-playbook --encrypt
- D. ansible-galaxy encrypt

Answer: A

Explanation: ansible-vault encrypt converts a plaintext file into an encrypted vault file.

Q47. Which flag tells ansible-playbook to prompt for the vault password?

- A. --vault-prompt
- B. --ask-vault-pass
- C. --vault-password
- D. --secret

Answer: B

Explanation: Use --ask-vault-pass to be prompted interactively, or --vault-password-file for non-interactive runs.

Q48. Which option simulates a playbook run without making changes?

- A. --dry-run
- B. --check
- C. --simulate
- D. --no-op

Answer: B

Explanation: --check enables Ansible's check mode (dry run). Use with --diff to also see file change previews.

Q49. Which flag limits a playbook run to a subset of inventory hosts?

- A. --limit
- B. --hosts
- C. --filter
- D. --only

Answer: A

Explanation: --limit accepts a host pattern and restricts execution to matching hosts within the inventory.

Q50. Which command-line tool is used to look up module documentation locally?

- A. ansible-help
- B. ansible-doc
- C. ansible-info
- D. ansible-man

Answer: B

Explanation: ansible-doc shows documentation, parameters, and examples for the specified module/plugin.